

Hamiltonian Equations of Motion - Approved Segments

4 approved segments, in reading order

Chapter 8 Hamiltonian and Hamilton's Eqs of Motion

$$L = T - V = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$$

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} \bigg|_{\substack{\dot{q}_j \\ q_j}}$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \bigg|_{\substack{\dot{q}_j \\ q_j}}$$

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_\theta = l_z$$

$$L = L(r, \dot{r}, p_\theta)$$

~~Lagrange eqn~~

Legendre Transformation $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \rightarrow q_1, \dots, q_n, p_1, \dots, p_n$

$$f(x, y) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

In Thermodynamics

$$\text{1st Law} \quad dU = dQ + dW = T dS - p dV \quad \text{Fluid at constant } N$$

$$U = U(S, V) \quad \text{Internal Energy}$$

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$F = A = U - TS$ Helmholtz Free Energy

$$dA = dU - d(TS) = \cancel{T}ds - p dV - \cancel{T}ds - S dT = -S dT - p dV \Rightarrow A = A(T, V)$$

In general

$$f(x, y) = u dx + v dy \quad u = \frac{\partial f}{\partial x} \quad v = \frac{\partial f}{\partial y} \quad (x, y) \longrightarrow (u, y)$$

$$g = f - ux \Rightarrow dg = df - d(ux) = \cancel{u}dx + v dy - \cancel{u}dx - x du = -x du + v dy$$

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$$g = g(u, y)$$

let's look at $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)$ $\left(p_i = \frac{\partial L}{\partial \dot{q}_i} \right) \bigg|_{\substack{q_j \\ \dot{q}_j}}$

$$H = -\left(L - \sum_i p_i \dot{q}_i \right) \quad dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \cancel{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i} \right) + \sum_i \left(\cancel{p_i d\dot{q}_i} + \dot{q}_i dp_i \right)$$

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right) \Rightarrow H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$$

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$$\boxed{\begin{aligned} \frac{\partial H}{\partial p_i} &= \dot{q}_i & \frac{\partial H}{\partial q_i} &= -\frac{\partial L}{\partial q_i} \stackrel{\text{Lagrange Eqn}}{=} -\dot{p}_i & \frac{\partial H}{\partial t} &= -\frac{\partial L}{\partial t} \end{aligned}}$$

CANONICAL EQS OF MOTION